

Solved selected problems of Classical Electrodynamics - Hans Ohanian

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Chapter 6 - The Theory of Special Relativity

Problems

Solution. 4.

- (a) Let the time interval between the birth of the blob at Q and its arrival at P be Δt_0 .

The time it takes for the light to arrive to the Earth from the birth of the blob at Q is

$$\Delta t_Q = \frac{QO}{c}$$

On the hand, the blob moves a distance $QP = V\Delta t_0$ towards the earth if we assume the angle is small. So the time it takes for the light emitted at P to reach the earth, assuming we started measuring the time when the blob was born, is

$$\Delta t_P = \frac{QO - QP}{c} + \Delta t_0$$

Then the time interval between the reception at the Earth of the light emitted at Q and the light emitted at P is

$$\begin{aligned}\Delta t &= \Delta t_P - \Delta t_Q \\ &= \frac{QO - QP}{c} + \Delta t_0 - \frac{QO}{c} \\ &= \Delta t_0 - \frac{V\Delta t_0}{c} \\ &= (1 - V)\Delta t_0\end{aligned}$$

Where we assumed that $c = 1$.

- (b) We note that $\sin \theta' = PP'/OP$ but if we assume the angle is small then we have that

$$PP' = \theta' OP$$

So, if we assume that Q and P are not too far appart, then the apparent transverse speed is given by

$$\frac{PP'}{\Delta t} = \frac{\theta' OP}{(1 - V)\Delta t_0} = \frac{OP}{QP} \frac{\theta' V}{1 - V}$$

Where we used that $\Delta t_0 = QP/V$.

Let us assume that $OP = 5 \text{ Gly}$, $QP = 500 \text{ Mly}$, $V = 0.99c$ (99% of the speed of light) and $\theta' = 0.1^\circ = 0.0017 \text{ rad}$ then

$$\frac{PP'}{\Delta t} = \frac{5 \times 10^9}{1 \times 10^6} \frac{0.0017 \cdot 0.99}{1 - 0.99} = 1.68$$

□

Solution. 5.

(a) By integration we note that

$$\begin{aligned}\int_{P_0}^P dP &\leq c^2 \int_{\rho_0}^{\rho} d\rho \\ P - P_0 &\leq c^2(\rho - \rho_0) \\ P &\leq P_0 + c^2(\rho - \rho_0)\end{aligned}$$

(b) The pressure in neutron-star material is $P_0 = 1.22 \times 10^{34} \text{ g/cm} \cdot \text{s}^2$ at a density of $\rho_0 = 4.6 \times 10^{14} \text{ g/cm}^3$. Then, the pressure at a density $\rho = 10^{20} \text{ g/cm}^3$ has the following upper bound

$$\begin{aligned}P &\leq 1.22 \times 10^{34} + (10^{20} - 4.6 \times 10^{14}) \cdot (3 \times 10^{10})^2 \\ &\leq 9 \times 10^{40} \text{ g/cm} \cdot \text{s}^2\end{aligned}$$

If we set $\rho = 10^{30} \text{ g/cm}^3$ then P has the following upper bound

$$\begin{aligned}P &\leq 1.22 \times 10^{34} + (10^{30} - 4.6 \times 10^{14}) \cdot (3 \times 10^{10})^2 \\ &\leq 9 \times 10^{50} \text{ g/cm} \cdot \text{s}^2\end{aligned}$$

□

Solution. 10. To compute a'_x we need to take the derivative of u'_x with respect to t' as follows

$$\begin{aligned}
a'_x &= \frac{du'_x}{dt'} = \frac{d}{dt'} \left(\frac{u_x - V}{1 - u_x V} \right) \\
&= \frac{dt}{dt'} \frac{d}{dt} \left(\frac{u_x - V}{1 - u_x V} \right) \\
&= \frac{dt}{\gamma(dt - Vdx)} \frac{d}{dt} \left(\frac{u_x - V}{1 - u_x V} \right) \\
&= \frac{\sqrt{1 - V^2}}{1 - u_x V} \left(\frac{a_x}{1 - u_x V} + \frac{(u_x - V)a_x V}{(1 - u_x V)^2} \right) \\
&= a_x \frac{\sqrt{1 - V^2}}{1 - u_x V} \left(\frac{1 - u_x V + (u_x V - V^2)}{(1 - u_x V)^2} \right) \\
&= a_x \frac{(1 - V^2)^{3/2}}{(1 - u_x V)^3}
\end{aligned}$$

In the same way, for a'_y we have that

$$\begin{aligned}
a'_y &= \frac{du'_y}{dt'} = \frac{dt}{dt'} \frac{d}{dt} \left(u_y \frac{\sqrt{1 - V^2}}{1 - u_x V} \right) \\
&= \frac{\sqrt{1 - V^2}}{1 - u_x V} \left(\frac{a_y \sqrt{1 - V^2}}{1 - u_x V} + \frac{a_x V (u_y \sqrt{1 - V^2})}{(1 - u_x V)^2} \right) \\
&= \frac{a_y (1 - V^2)}{(1 - u_x V)^2} + \frac{a_x u_y V (1 - V^2)}{(1 - u_x V)^3}
\end{aligned}$$

And finally, for a'_z

$$\begin{aligned}
a'_z &= \frac{du'_z}{dt'} = \frac{dt}{dt'} \frac{d}{dt} \left(u_z \frac{\sqrt{1 - V^2}}{1 - u_x V} \right) \\
&= \frac{\sqrt{1 - V^2}}{1 - u_x V} \left(\frac{a_z \sqrt{1 - V^2}}{1 - u_x V} + \frac{a_x V (u_z \sqrt{1 - V^2})}{(1 - u_x V)^2} \right) \\
&= \frac{a_z (1 - V^2)}{(1 - u_x V)^2} + \frac{a_x u_z V (1 - V^2)}{(1 - u_x V)^3}
\end{aligned}$$

□

Solution. 11.

(a) We know from problem 10 that

$$a_x = (1 - V^2)^{3/2} \frac{a'_x}{(1 - Vu'_x)^3}$$

Since at the frame of the spaceship $u'_x = 0$ and $a_x = dV/dt$ then

$$\frac{dV}{dt} = (1 - V^2)^{3/2} a'_x$$

So by integration we have that

$$\begin{aligned} \int_0^V (1 - V^2)^{-3/2} dV &= \int_0^t a'_x dt \\ \frac{V}{\sqrt{1 - V^2}} &= a'_x t \end{aligned}$$

Also, since $u'_x = 0$ we have that $u_x = V$ and hence

$$\begin{aligned} \frac{u_x}{\sqrt{1 - u_x^2}} &= a'_x t \\ u_x^2 &= (a'_x t)^2 (1 - u_x^2) \\ u_x^2 (1 + (a'_x t)^2) &= (a'_x t)^2 \\ u_x^2 &= \frac{(a'_x t)^2}{1 + (a'_x t)^2} \\ \frac{dx}{dt} &= \frac{a'_x t}{\sqrt{1 + (a'_x t)^2}} \end{aligned}$$

Then by integration again we have that

$$\begin{aligned} \int_0^x dx &= a'_x \int_0^t \frac{t}{\sqrt{1 + (a'_x t)^2}} dt \\ x &= a'_x \left[\frac{\sqrt{1 + (a'_x t)^2}}{(a'_x)^2} - \frac{1}{(a'_x)^2} \right] \\ x &= \frac{1}{a'_x} [\sqrt{1 + (a'_x t)^2} - 1] \end{aligned}$$

(b) When $V \ll 1$ we see that

$$a'_x t = \frac{V}{\sqrt{1 - V^2}} \approx V$$

So $a'_x t \ll 1$. Then, we can approximate the worldline equation using the binomial approximation as follows

$$x \approx \frac{1}{a'_x} \left[1 + \frac{(a'_x t)^2}{2} - 1 \right] = \frac{1}{2} a'_x t^2$$

□

Solution. 12.

- (a) Let a particle move with speed u' along a straight line in the $x'-y'$ plane, the direction of the motion of the particle makes an angle θ' with the x' axis.

The components of the velocity in each direction in the $x'y'z'$ reference frame are

$$u'_x = u' \cos \theta' \quad u'_y = u' \sin \theta'$$

Then, the velocity components in the xyz reference frame are

$$u_x = \frac{u'_x + V}{1 + u'_x V} = \frac{u' \cos \theta' + V}{1 + u' \cos \theta' V}$$
$$u_y = \frac{u'_y \sqrt{1 - V^2}}{1 + u'_x V} = \frac{u' \sin \theta' \sqrt{1 - V^2}}{1 + u' \cos \theta' V}$$

Also, u_x and u_y are related by $\tan \theta = u_y/u_x$ where θ is the angle the velocity u makes with the x -axis, so

$$\tan \theta = \frac{u_y}{u_x} = \frac{\frac{u' \sin \theta' \sqrt{1 - V^2}}{1 + u' \cos \theta' V}}{\frac{u' \cos \theta' + V}{1 + u' \cos \theta' V}} = \frac{u' \sin \theta' \sqrt{1 - V^2}}{u' \cos \theta' + V}$$

- (b) Suppose that a source at the origin of the $x'y'z'$ reference frame emits particles in all directions.

If $V \rightarrow 1$ then for any angle $\theta' \in [0, 2\pi]$ we see that

$$\lim_{V \rightarrow 1} \tan \theta = \lim_{V \rightarrow 1} \frac{u' \sin \theta' \sqrt{1 - V^2}}{u' \cos \theta' + V} = 0$$

This is because $\sin \theta'$ becomes 0 at $\theta' = 0, \pi$ and 2π , making the limit 0 but the denominator is never 0 since $V \rightarrow 1$.

If $\sin \theta' \neq 0$ the limit also goes to 0 since $\sqrt{1 - V^2} \rightarrow 0$ as $V \rightarrow 1$.

So in any case the limit goes to 0.

Therefore, must be that $\theta = \pi n$ where $n \in \mathbb{Z}$, so from the xyz reference frame the particles form two narrow beams in the forward direction ($\theta = 0, 2\pi, 4\pi, \dots$) and in the backward direction ($\theta = \pi, 3\pi, 5\pi, \dots$).

□

Solution. 19. Let a spaceship leave Earth on a excursion to Barnard's star. The position of the spaceship as a function of time is

$$x = \frac{c^2}{g} \left(\sqrt{1 + \frac{g^2 t^2}{c^2}} - 1 \right)$$

Where t is the time measured in the reference frame of the Earth, with $t = 0$ at the start of the trip, and $g = 981 \text{ cm/s}^2$ is the acceleration of the spaceship in its own reference frame.

(a) Taking the time derivative of x we have that

$$v = \frac{dx}{dt} = \frac{c^2}{g} \frac{g^2 t}{c^2 \sqrt{1 + \frac{g^2 t^2}{c^2}}} = \frac{gt}{\sqrt{1 + \frac{g^2 t^2}{c^2}}}$$

(b) When 2 years have elapsed on Earth, then the moving clock on the spaceship has elapsed

$$\begin{aligned} \tau &= \int_0^{2 \text{ y}} \sqrt{1 - v^2/c^2} \, dt \\ \tau &= \int_0^{2 \text{ y}} \sqrt{1 - \frac{g^2 t^2}{c^2 + g^2 t^2}} \, dt \\ \tau &= \int_0^{2 \text{ y}} \sqrt{\frac{c^2 + g^2 t^2 - g^2 t^2}{c^2 + g^2 t^2}} \, dt \\ \tau &= \int_0^{2 \text{ y}} \frac{1}{\sqrt{1 + g^2 t^2/c^2}} \, dt \\ \tau &= \int_0^{6.3 \times 10^7 \text{ s}} \frac{1}{\sqrt{1 + (981)^2 t^2 / (3 \times 10^{10})^2}} \, dt \\ \tau &= 4.5 \times 10^7 \text{ s} \approx 1.42 \text{ years} \end{aligned}$$

(c) When 2 years have been elapsed in the Earth then the position of the spaceship is

$$x = \frac{(3 \times 10^{10})^2}{981} \left(\sqrt{1 + \frac{(981)^2 (6.3 \times 10^7)^2}{(3 \times 10^{10})^2}} - 1 \right) = 1.183 \times 10^{18} \text{ cm}$$

The light needs to travel back to Earth the distance we just computed and since the speed of light is the same for any observer, the time it takes for the radio signal to reach the Earth is

$$t = \frac{x}{c} = \frac{1.183 \times 10^{18}}{3 \times 10^{10}} = 39133333.3 \text{ s} = 1.24 \text{ years}$$

- (d) We know that the velocity of the spaceship at the moment of sending the radio signal is

$$v = \frac{981 \cdot 6.3 \times 10^7}{\sqrt{1 + \frac{(981)^2 (6.3 \times 10^7)^2}{(3 \times 10^{10})^2}}} = 26988436025.48 \text{ cm/s}$$

Since the signal is emitted with a frequency of $1 \times 10^9 \text{ Hz}$ in the reference frame of the spaceship (the source), then the frequency at the arrival on Earth will be

$$\nu = \nu_0 \sqrt{1 - v^2/c^2} = 1 \times 10^9 \sqrt{1 - (0.899)^2} = 4.37 \times 10^8 \text{ Hz}$$

□

Solution. 20.

We see that the instantaneous velocities of each clock are

$$V_1(t) = \frac{dx}{dt} = V_0 \quad \text{and} \quad V_2(t) = \frac{dx}{dt} = a_0 t$$

Then the proper time on the first clock (following a straight worldline) is

$$\Delta\tau_1 = \int_{t_A}^{t_B} \sqrt{1 - V_0^2} \, dt = t_B \sqrt{1 - V_0^2}$$

Where we used that $t_A = 0$.

For the second clock, following a curved worldline we have that

$$\begin{aligned} \Delta\tau_2 &= \int_{t_A}^{t_B} \sqrt{1 - a_0^2 t^2} \, dt \\ &= \left[\frac{a_0 t \sqrt{1 - a_0^2 t^2} + \arcsin(a_0 t)}{2a_0} \right]_0^{t_B} \\ &= \frac{1}{2} t_B \sqrt{1 - a_0^2 t_B^2} + \frac{\arcsin(a_0 t_B)}{2a_0} \end{aligned}$$

We notice that when $t = t_B$ we have that $x_B = V_0 t_B$ and $x_B = \frac{1}{2} a_0 t_B^2$ so

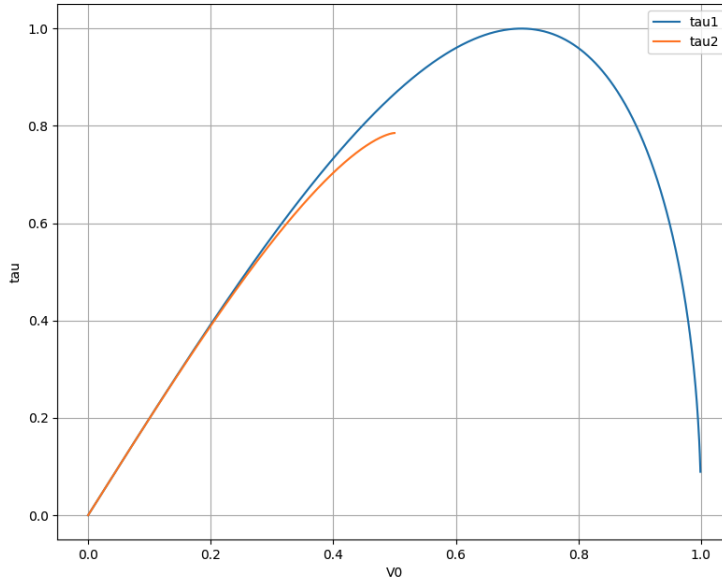
$$\begin{aligned} \frac{1}{2} a_0 t_B^2 &= V_0 t_B \\ t_B &= \frac{2V_0}{a_0} \end{aligned}$$

Where we used that the worldlines have the same value at $t = t_B$.

Then the proper time equations become

$$\Delta\tau_1 = \frac{2V_0}{a_0} \sqrt{1 - V_0^2} \quad \Delta\tau_2 = \frac{V_0}{a_0} \sqrt{1 - 4V_0^2} + \frac{\arcsin(2V_0)}{2a_0}$$

Below, we plotted the real part of both equations for different values of V_0



We notice that $\Delta\tau_1$ is always bigger than $\Delta\tau_2$ so the clock following the straight worldline shows the longer proper time. $\square$

Solution. 21.

Let us consider a reference frame in which the free particle is at rest, then in this reference frame from the event A to the event B the proper time is

$$\Delta\tau_1 = \int_{t_A}^{t_B} \sqrt{1 - 0^2} \, dt = \int_{t_A}^{t_B} dt = \Delta t$$

From this reference frame, taking a spacetime diagram in which the position of the particle matches the origin, the worldline of the particle is a line along the t -axis. So any other worldline between the events A and B must be a line in which the particle is not free, since it needs to move away from the t -axis and then come back. For such a worldline the proper time is

$$\Delta\tau_2 = \int_{t_A}^{t_B} \sqrt{1 - V(t)^2} \, dt$$

Since $V(t) \leq 1$ then $\sqrt{1 - V(t)^2} \leq 1$. Also, in between the events A and B there must be an interval where $V(t) \neq 0$ otherwise we would be following the free particle worldline, suppose between the events C and D we have that $V(t) \neq 0$ then we have that $\sqrt{1 - V(t)^2} < 1$ in that interval.

Therefore

$$\Delta\tau_2 = \int_{t_A}^{t_B} \sqrt{1 - V(t)^2} \, dt < \int_{t_A}^{t_B} dt = \Delta\tau_1$$

Hence, the proper time of a free particle is the largest. □

Solution. 22.

Let us consider the star Capella at a distance of 46 light-years from Earth. An astronaut wants to travel from Earth to Capella in a time no more than 20 years as reckoned by clocks aboard his spaceship. Seen from Earth the astronaut moves with a velocity

$$V = \frac{\Delta x}{\Delta t}$$

We want that the clock aboard the spaceship measures at the end of the trip $\Delta\tau = 20$ years and we know that $\Delta\tau$ and Δt are related by

$$\Delta t = \frac{\Delta\tau}{\sqrt{1 - V^2}}$$

So the spaceship needs to travel at a speed of

$$\begin{aligned} V &= \frac{\Delta x}{\Delta t} = \frac{\Delta x \sqrt{1 - V^2}}{\Delta\tau} \\ V^2 &= \frac{\Delta x^2}{\Delta\tau^2} (1 - V^2) \\ V^2 + \frac{\Delta x^2}{\Delta\tau^2} V^2 &= \frac{\Delta x^2}{\Delta\tau^2} \\ V &= \frac{\Delta x}{\Delta\tau} \left(1 + \frac{\Delta x^2}{\Delta\tau^2} \right)^{-1/2} \end{aligned}$$

Replacing values

$$\begin{aligned} V &= \frac{46 \text{ light-years}}{20 \text{ years}} \left(1 + \frac{(46 \text{ light-years})^2}{(20 \text{ years})^2} \right)^{-1/2} \\ &= 0.917 \, c \end{aligned}$$

Finally, a clock on Earth would measure

$$\Delta t = \frac{20 \text{ years}}{\sqrt{1 - (0.917)^2}} = 50.13 \text{ years}$$

□

Solution. 25.

We know that the Earth moves around the Sun at a speed of 30 km/s , if we assume that the Earth is a perfect sphere of radius $6.4 \times 10^6 \text{ m}$ then the east-west diameter will be contracted by

$$2R - 2R\sqrt{1 - \frac{V^2}{c^2}} = 2 \cdot 6400 \cdot \left(1 - \sqrt{1 - \frac{30^2}{300000^2}}\right) = 0.064 \text{ m}$$

The length contraction is barely noticeable since the velocity of the Earth is 0.01% of the speed of light.

The transverse directions are not length contracted so, if the Earth has a volume of

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi 6400^3 = 1.09 \times 10^{12} \text{ km}^3$$

Then from the sun the volume will be contracted by

$$V - V\sqrt{1 - \frac{V^2}{c^2}} = 1.09 \times 10^{12} \cdot \left(1 - \sqrt{1 - \frac{30^2}{300000^2}}\right) = 5450 \text{ km}^3$$

□